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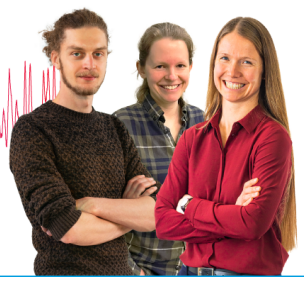
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Spectroscopic study of the $C\,^1\Sigma^+$ state of ^{7}LiH

Jye-Jong Chen and Wei-Tzou Luh[a)]
Department of Chemistry, National Chung Hsing University, 250 Kuo-Kuang Road, Taichung 402, Taiwan, Republic of China

Gwang-Hi Jeung
Laboratoire Aimé Cotton (CNRS UPR-3321), Bât 505, and Applications Scientifiques du Calcul Intensif (CNRS UPR-9029), Bât 506, Universté de Paris-Sud, 91405 Orsay, France

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Forty-two vibrational ($v=2-43$) levels of the ^{7}LiH $C\,^1\Sigma^+$ excited electronic state have been observed by a pulsed optical–optical double resonance fluorescence depletion spectroscopic technique. The absolute vibrational numbering of the $C\,^1\Sigma^+$ state is identified with the measurements of the isotopic shifts between ^{7}LiH and ^{6}LiH among several rotation-vibration bands of the $C\,^1\Sigma^+-A\,^1\Sigma^+$ electronic system, and it is further demonstrated by the consistency between observed spectral intensities and calculated ones. The highest observed quasi bound rotational level for the $v=43$ level is $J=8$. The spectral term values for those vibrational levels lying above $v=33$ occur in an irregular order. The level of $v=34$ lies mainly in the inner, tiny well. The term values for the observed rovibrational levels ($2\leqslant v\leqslant 32, 0\leqslant J\leqslant 11$) in the outer ionic well are described by a set of Dunham-type coefficients, with which a Rydberg–Klein–Rees (RKR) potential energy curve is constructed. This RKR potential is combined with an *ab initio* potential to construct an effective hybrid potential. The calculated electronic transition moment function for the *C*-*A* transition is also given. © *1999 American Institute of Physics.* [S0021-9606(99)00209-3]

I. INTRODUCTION

Lithium hydride, the simplest neutral heteronuclear molecule, has been the object of intensive theoretical and spectroscopic interest since the 1930s. A comprehensive and critical review for this molecule has been recently compiled by Stwalley and Zemke.[1] However, up to now only the lowest three electronic states, $X\,^1\Sigma^+$, $A\,^1\Sigma^+$, and $B\,^1\Pi$, have been identified and well characterized by conventional spectroscopy[2–7] and by laser spectroscopy,[8–16] although some other electronic states have been predicted theoretically for almost three decades.[17–22] In a recent effective core potential calculation, Boutalib and Gadéa[22] reported that the potential energy curves for highly excited electronic states of LiH would become complex due to the avoided crossings between the valence and Rydberg atomic configurations on one hand and between the ionic and neutral configurations on the other hand. For the intriguing double-well $C\,^1\Sigma^+$ electronic state, in a preliminary account, Lin *et al.*[23] reported their first spectroscopic observation using a pulsed optical–optical double resonance (OODR) fluorescence depletion spectroscopic technique.

In this work, we have extended our observation of the $C\,^1\Sigma^+$ state of ^{7}LiH from its third vibrational level $v=2$ to its upmost vibrational level $v=43$. The absolute vibrational assignment of the $C\,^1\Sigma^+$ state is identified with the measurements of the isotopic shifts between ^{6}LiH and ^{7}LiH among several rotation-vibration bands of the $C\,^1\Sigma^+-A\,^1\Sigma^+$ electronic system. For the rovibrational levels in the outer ionic well, a set of Dunham-type coefficients and a related RKR potential energy curve are given. An *ab initio* potential energy curve for the $C\,^1\Sigma^+$ state and a transition moment function for the $C\,^1\Sigma^+-A\,^1\Sigma^+$ electronic system are calculated. An effective hybrid potential is also constructed for the $C\,^1\Sigma^+$ state.

II. EXPERIMENT

Figure 1 portrays the experimental setup, which was renovated from our preliminary setup.[23] This renovated setup had a better sensitivity and it allowed us to observe some weak ^{6}LiH OODR signals, significant for absolute vibrational assignment. In this work, two different laser systems were used. The first laser system was composed of one frequency-doubled 10-Hz *Q*-switched Nd:YAG laser (Quantel BW), with a pulse width of 4 ns and at a pulse energy of 30 mJ, and one dye laser (Lumonics HD 500). The dye laser beam was frequency-doubled by a frequency-doubling system (Inrad autotracker II) to generate an ultraviolet PUMP laser beam for exciting the LiH molecule from its $X\,^1\Sigma^+$ state to its $A\,^1\Sigma^+$ state. For covering the optimal spectral range of four LiH $A(6-9)-X(0)$ bands, the methanol solutions of two laser dyes, LDS698 and LDS722 (Exciton), were used as active media. The second laser system was composed of another frequency-doubled 10-Hz *Q*-switched Nd:YAG laser (Lumonics YM 600), with a pulse width of 7 ns and at a pulse energy of 20 mJ, and another dye laser (Lumonics HD 500). The dye laser beam was used as a PROBE laser beam for probing the excited electronic state of

a)Author to whom correspondence should be addressed; electronic mail: wtluh@dragon.nchu.edu.tw

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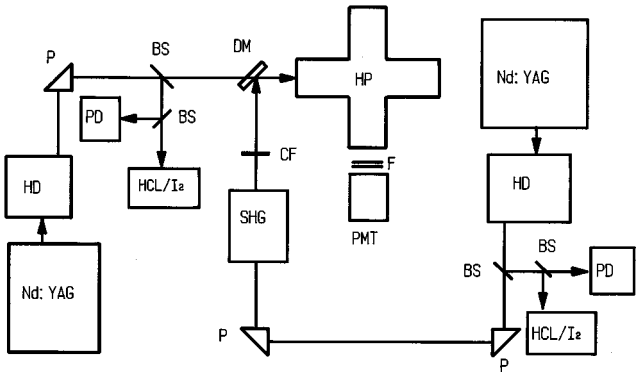


FIG. 1. Experimental schematics. BS: a beam splitter; DL: a dye laser; DM: a dichroic mirror; F: an optical filter; FD: a frequency-doubling system; H: a hollow cathode lamp; HP: a lithium heat-pipe oven; I: an iodine cell; L: a focusing lens; Nd: a Nd:YAG laser; P: a 90° prism; PM: a photomultiplier tube.

interest. The methanol solutions of 11 laser dyes, R575, R590, R610, R640, DCM, LDS698, LDS722, LDS759, LDS821, LDS867, and LDS925, were used as active media. Note that we were not able to observe the lowest two vibrational levels $v=0$ and 1 because the effective longest wavelength for our dye laser is 965 nm. The laser pulse energy and the laser linewidth were, respectively, 0.1–0.2 mJ and 0.1 cm^{-1} for the PUMP laser, and 0.3–0.5 mJ and 0.04 cm^{-1} for the PROBE laser. The laser frequencies above 14 200 cm^{-1} were calibrated by a reference spectrum of I_2, and those below 14 200 cm^{-1} were calibrated by an optogalvanic spectrum, taken either with a Ne-filled sodium/potassium hollow cathode lamp (Perkin Elmer) or with an Ar-filled lithium hollow cathode lamp (Perkin Elmer). The uncertainty of frequency measurements is about 0.1 cm^{-1}. The PUMP laser beam and the PROBE laser beam were aligned collinearly along the axis of the longer arms of a five-arm crossed heat-pipe oven. The pulse delay between the PROBE laser pulse and the PUMP laser pulse was regulated by two digital delay/pulse generators (Stanford DG 535). The heat-pipe oven, containing 10 g of naturally abundant lithium metal (STREM, 3*N*) in its center, was normally operated under 1 Torr of Ar gas and at the temperature 953 K. At 953 K, LiH molecules most likely populate the rotational level $J''=6$. The induced $A-X$ fluorescence was detected by a photomultiplier (EMI 9863QB) detector after a spectral bandpass filter. The detector signal was fed into a boxcar-averager-gated-integrator system, and the averaged output was then recorded by a personal computer. The OODR fluorescence depletion spectrum was obtained by monitoring the variation of the induced $A-X$ fluorescence intensity as the PROBE laser was scanned, while the PUMP laser was fixed to a particular LiH $A-X$ rovibrational transition.

III. CALCULATION

For the $C\,^1\Sigma^+$ state, Boutalib and Gadéa[22] reported only one effective core potential (ECP) energy for an internuclear distance longer than 20 bohr, and in turn their potential could not adequately describe those vibrational levels near the dissociation limit (see Table VI below). Furthermore, there is also no transition moment function available for the $C\,^1\Sigma^+ - A\,^1\Sigma^+$ electronic transition. Thus a further calculation on relevant potential energy curves and transition moments is desirable.

A large set of the atomic basis function was used for the hydrogen and lithium atoms. Each Gaussian atomic primitive function was first optimized to best describe the lowest electronic states of different symmetry (^2S, ^2P, ^2D, and ^2F). For higher atomic states of each symmetry, additional diffuse Gaussian functions were chosen in a geometric progression. For the high angular momentum of the hydrogen basis, the Gaussian exponents minimizing both the anion energy and the interatomic polarization-correlation energies were obtained. The resultant basis sets have been reported elsewhere.[24] For the atomic ground state, our Hartree–Fock (HF) energies (in hartrees) are $-0.499\,71$ for H and $-7.432\,57$ for Li, which are close to their HF limits,[25] -0.5 and $-7.432\,73$, respectively; and our configuration-interaction (CI) energy for Li is -203.344 eV, which is close to the experimental value,[26] -203.429 eV. The nearly full CI ionization energies of the lithium atom from its ground and excited electronic states were in good agreement with experimental data. The calculated electron affinity of the hydrogen atom is 0.736 eV, which is close to the experimental value,[27] 0.756 eV. Note that a good description of the hydrogen electron affinity was found to be rather crucial to obtain reliable potential energy curves, particularly the ionic part, for the $^1\Sigma^+$ states of LiH.

For molecular configuration interaction (CI) calculations, we used the MOLCAS program package.[28] For LiH, all electrons were allowed to occupy a large set of active space and all possible single and double substitutions were included. This led to a full CI quality calculation. The molecular orbital basis was optimized by doing the multiconfiguration HF, although the choice of particular molecular orbital sets was demonstrated not to affect the final CI energies, indicating indirectly that our calculation is of the nearly full CI quality. Our calculation is fully *ab initio* since all electrons are explicitly and extensively described and no approximation (except a finite basis set and a finite number of configurations) is made.

We have calculated the potential energy curves for several $^1\Sigma^+$ electronic states[24] and the transition moment functions for several electronic transitions. Table I presents only the relevant results for the potential energy curve of the $C\,^1\Sigma^+$ electronic state and the transition moment function for the $C\,^1\Sigma^+ - A\,^1\Sigma^+$ electronic transition, which are discussed further in Sec. IV.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. OODR results

The OODR spectra were taken by monitoring the induced ^{7}LiH $A\,^1\Sigma^+ - X\,^1\Sigma^+$ fluorescence intensity at 379 nm through a set of optical filters, having a spectral bandpass [full width at half maximum (FWHM)] of 36 nm. Several rovibrational energy levels ($v'=6-9, J'=0-10$) of the $A\,^1\Sigma^+$ state were used as intermediate states. We observed a total of 734 transitions, corresponding to 458 rovibrational

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TABLE I. Theoretical calculations of the $C\,^1\Sigma^+$ potential energy curve $U(R)$ [hartree] and the transition moment function $\mu(R)$ [a.u.] for the $C\,^1\Sigma^+ - A\,^1\Sigma^+$ electronic transition. R is the internuclear distance (bohr).

R	$U(R)$[a]	$\mu(R)$
1.75	0.148 900	0.1798
2.0	0.087 909	0.1603
2.5	0.025 698	0.0542
2.75	0.011 022	0.0151
3.0	0.002 487	0.0687
3.25	−0.002 351	0.1430
3.5	−0.004 576	0.2095
3.75	−0.005 509	0.2824
4.0	−0.005 354	0.3552
4.5	−0.004 200	0.6559
5.0	−0.003 317	1.5147
5.5	−0.005 792	2.6906
6.0	−0.011 074	3.1122
7.0	−0.022 364	3.6695
8.0	−0.030 898	4.1036
9.0	−0.036 012	4.5623
10.0	−0.038 043	4.9433
11.0	−0.037 156	4.7118
12.0	−0.033 983	3.7846
13.0	−0.029 682	2.8098
15.0	−0.020 847	1.6204
17.0	−0.013 210	1.1004
19.0	−0.006 937	0.9336
20.0	−0.004 172	1.0739
21.0	−0.001 912	1.2667
22.0	−0.000 636	1.6285
23.0	−0.000 177	1.7493
25.0	−0.000 030	1.7433
28.0	−0.000 003	1.7261
30.0	−0.000 001	1.7269

[a]With respect to the dissociation limit of $\mathrm{Li}(3s)+\mathrm{H}(1s)$.

levels among 42 vibrational levels of the ^{7}LiH $C\,^1\Sigma^+$ state. Figure 2 presents the data field measured. Our lowest and highest observed vibrational levels are, respectively, the (v_0-10) level and the $(31+v_0)$ level, corresponding to the two exact vibrational quantum numbers 2 and 43 after the reference level v_0 was identified as 12 by isotopic shift measurements (which are detailed in the next section). We could not detect the lowest two vibrational levels, $v=0$ and $v=1$, because their expected transition lines, having favorable Franck–Condon factors and transition moments, are in the spectral range 1.1–1.3 μm, which lies outside the effective wavelength limit of 965 nm for the dye laser we had. For the rovibrational levels above the vibrational level $v=33$, their term values were found to occur in an irregular order, particularly in the range of $v=34-38$, as shown in Fig. 3. Note that in Fig. 3 the calculated term values (in open triangles) from an effective hybrid potential (see Sec. IV C below) are also presented, and the solid connecting lines between observed term values are only for the purpose of visualization. We could not observe the lowest two rotational levels, (34,0) and (34,1), of the vibrational level $v=34$, which was theoretically predicted to be trapped in an inner tiny neutral well[22] and will be further discussed below. The highest observed vibrational state is $v=43$. For the spectral range beyond the depletion peak to the levels $(43, J\leqslant 8)$, there appear weak, broad depletion transitions, as demonstrated in

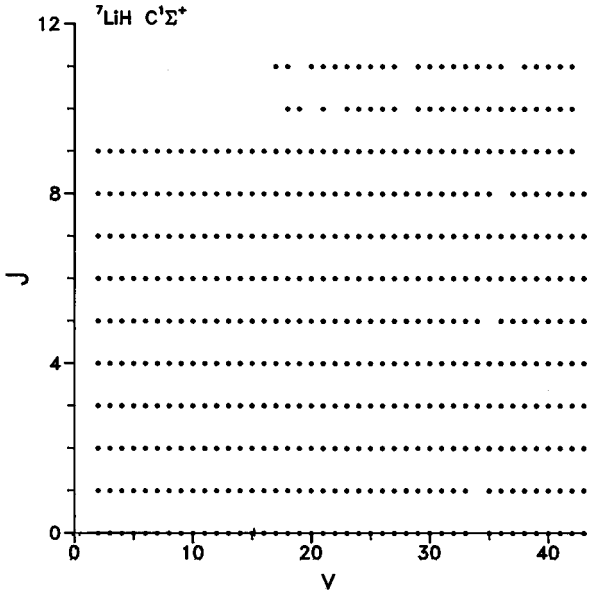


FIG. 2. The (v,J) data field observed. A total of 458 rovibrational levels of the $C\,^1\Sigma^+$ electronic state.

Fig. 4, in which two observed spectra are reproduced and each spectrum is composed of three bound-to-bound transitions on the lower frequency side and weak, broad depletion transitions on the higher frequency side, and the vertical dashed line denotes the rotationless dissociation limit of the $C\,^1\Sigma^+$ state, beyond which C-A free-bound transitions may come into play.

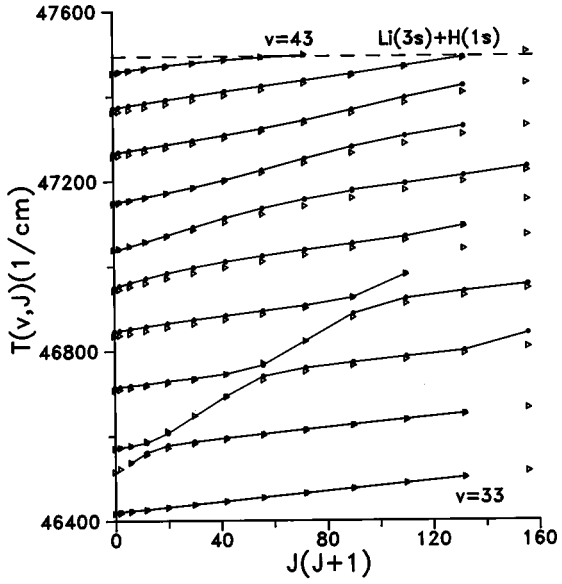


FIG. 3. A plot of the observed term energies $T(v,J)$ with respect to $J(J+1)$ for the 124 observed rovibrational levels, among the 11 vibrational levels, $v=33-43$, of the LiH $C\,^1\Sigma^+$ electronic state. The upper, horizontal dashed line indicates the dissociation limit, $\mathrm{Li}(3s)+\mathrm{H}(1s)$. Open triangles denote the term values calculated from the CLJ-RKR-CLJ hybrid potential (see Sec. IV C and Table V.)

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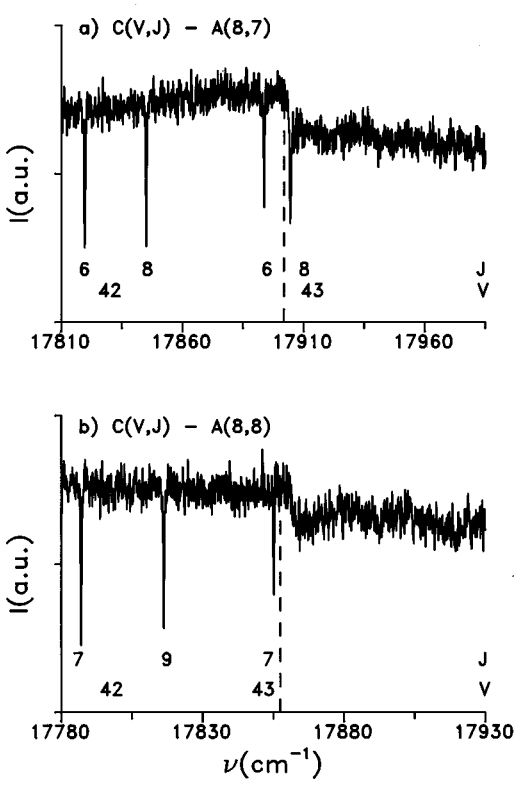


FIG. 4. OODR fluorescence depletion spectra of ^{7}LiH. Spectrum (a) portrays the $P(7)$ and $R(7)$ OODR transitions from the intermediate $A(8,7)$ level to the vibrational levels 42 and 43 of the $C\,^1\Sigma^+$ electronic state, and spectrum (b) the $P(8)$ and $R(8)$ OODR transitions from the intermediate $A(8,8)$ level to the vibrational levels 42 and 43. The vertical dashed line in each spectrum indicates the hypothetical rotationless dissociation limit from the intermediate level.

B. Absolute vibrational numbering and molecular constants

To absolutely identify the vibrational quantum number of the ^{7}LiH $C\,^1\Sigma^+$ state, one may measure the crucial isotopic shifts[29] between ^{7}LiH and one of its isotopic molecules among some pertinent rotation-vibration bands of the $C\,^1\Sigma^+ - A\,^1\Sigma^+$ electronic system. Since there exist three other natural isotopic lithium hydride molecules, ^{6}LiH, ^{7}LiD, and ^{6}LiD, one may select one of them for isotopic shift measurements. In the heat-pipe oven we used, two isotopic molecules, ^{7}LiH and ^{6}LiH, would occur simultaneously in a relative ratio of ~12.33/1, according to their natural abundance of 92.5% ^{7}Li and 7.5% ^{6}Li. For some strong ^{7}LiH C-A transitions, we measured their respective ^{6}LiH C-A transitions. The results are partly presented in Table II, in which each $\delta\nu$ value in the last three columns represents a deviation of an expected isotopic shift $\Delta\nu$ from its respective observed isotopic shift $\Delta\nu_{\rm obs}$. The expected isotopic shift $\Delta\nu$ was calculated in the following manner.

First, the quantum number v_0 of the reference level was assumed as one likely value, e.g., 12, then the observed term values in the vibrational range, $(v_0-10)-(20+v_0)$, of the ^{7}LiH $C\,^1\Sigma^+$ state were fitted into a set of Dunham-type coefficients. The derived Dunham-type coefficients were then combined with the known constants[30] for the $A\,^1\Sigma^+$ state to calculate the expected isotopic shifts $\Delta\nu$ for some pertinent

TABLE II. Observed isotopic shifts $\Delta\nu_{\rm obs}$ between ten $C\,^1\Sigma^+(v+v_0,J) - A\,^1\Sigma^+(v',J')$ electronic transitions of ^{6}LiH and those of ^{7}LiH, and calculated differences $\delta\nu$ for three different absolute vibrational assignments ($v_0=11$, 12, and 13). $(v+v_0,J)$ represents the observed upper level $C(v+v_0,J)$; (v',J') the intermediate level $A(v',J')$; ν_6 the observed C-A transition frequency of ^{6}LiH; ν_7 the observed C-A transition frequency of ^{7}LiH; $\delta\nu_{11}$ the deviation of the expected isotopic shift $\Delta\nu_{11}$ (see also the text) from the observed isotopic shift $\Delta\nu_{\rm obs}$ if v_0 is assigned to be 11; $\delta\nu_{12}$ if v_0 is 12; and $\delta\nu_{13}$ if v_0 is 13. All values listed in the last six columns are given in cm^{-1}.

$(v+v_0,J)$	(v',J')	ν_6	ν_7	$\Delta\nu_{\rm obs}$	$\delta\nu_{11}$	$\delta\nu_{12}$	$\delta\nu_{13}$
$(12+v_0,5)$	(8,4)	15 435.49	15 423.50	11.99	−1.50	0.44	2.37
$(12+v_0,5)$	(8,6)	15 372.48	15 361.66	10.82	−1.54	0.39	2.32
$(13+v_0,5)$	(8,4)	15 621.10	15 608.26	12.84	−1.39	0.49	2.39
$(13+v_0,5)$	(8,6)	15 558.18	15 546.48	11.70	−1.47	0.42	2.31
$(14+v_0,5)$	(8,4)	15 802.89	15 789.14	13.75	−1.44	0.42	2.28
$(14+v_0,5)$	(8,6)	15 739.87	15 727.31	12.56	−1.46	0.40	2.25
$(16+v_0,5)$	(9,4)	15 760.72	15 749.69	11.03	−1.38	0.41	2.18
$(16+v_0,5)$	(9,6)	15 698.43	15 688.62	9.81	−1.35	0.43	2.20
$(17+v_0,5)$	(9,4)	15 930.88	15 919.33	11.55	−1.23	0.51	2.23
$(17+v_0,5)$	(9,6)	15 868.60	15 858.23	10.37	−1.24	0.50	2.21

C-A transitions by using the well-known relations.[29] We have tested five different v_0 values, 10–14, and the smallest three deviations, $\delta\nu_{11}$, $\delta\nu_{12}$, and $\delta\nu_{13}$, are presented in Table II. One can tell from Table II that the minimum deviation is $\delta\nu_{12}$, which in turn indicates that the true quantum number v_0 is 12, as was predicted previously in our preliminary account.[23] This assignment can be further demonstrated by the consistency between observed C-A OODR intensities and relevant calculated intensities (to be discussed below). The best set of Dunham-type coefficients, generated from a least-squared fit of the observed 451 term values, corresponding to 297 rovibrational levels among the vibrational range, 2–32, are presented in Table III, in which the Dunham correction, Y_{00}, its dissociation energy, D_e, and its electronic term value, T_e, are also given. The overall standard error is 0.045 cm^{-1}.

C. Potential energy curves

The Dunham-type coefficients listed in Table III are used to construct an RKR potential energy curve by using

TABLE III. Dunham-type coefficients Y_{ij} and some important spectroscopic parameters for the LiH $C\,^1\Sigma^+$ state. All values are in cm^{-1}. Each parenthesized number represents one standard deviation for the last few digits. The overall standard error is 0.045 cm^{-1}.

$Y_{10}=295.492\,881\,1(17\,932\,4)$
$Y_{20}=-2.231\,327\,031(2\,795\,804)$
$10^2Y_{30}=-1.397\,144\,51(19\,161\,98)$
$10^4Y_{40}=6.618\,843\,139(59\,357\,126)$
$10^6Y_{50}=-7.302\,975\,532(68\,099\,067)$
$Y_{01}=0.664\,333\,4826(569\,937\,4)$
$10^3Y_{11}=4.204\,277\,746(126\,124\,299)$
$10^4Y_{21}=-4.363\,117\,924(78\,504\,084)$
$10^6Y_{31}=8.170\,035\,739(144\,154\,404)$
$10^5Y_{02}=-4.928\,631\,935(212\,141\,952)$
$Y_{00}=-0.511$
$D_e=8465.31(\pm0.3)$
$T_e=39\,028.26(\pm0.3)$

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TABLE IV. RKR potential energy curve derived from Dunham-type coefficients of the ^{7}LiH $C\,^1\Sigma^+$ state. v is the vibrational quantum number; the rotational constant, B_v, and the corrected vibrational energy, G_v+Y_{00}, are in cm^{-1}; inner and outer turning points, $R_{\min}$ and $R_{\max}$, are in angstroms. Note that the levels $v=0$ and 1 are not observed, and the equilibrium internuclear distance, R_e, is 5.3660 Å.

v	B_v	G_v+Y_{00}	$R_{\min}$	$R_{\max}$
0	0.666 327	146.67	5.0124	5.7331
1	0.669 685	437.66	4.7591	6.0169
2	0.672 244	724.08	4.5881	6.2232
3	0.674 053	1005.87	4.4516	6.3996
4	0.675 161	1283.00	4.3355	6.5599
5	0.675 617	1555.44	4.2336	6.7103
6	0.675 470	1823.17	4.1421	6.8542
7	0.674 769	2086.20	4.0590	6.9937
8	0.673 563	2344.51	3.9825	7.1302
9	0.671 901	2598.14	3.9116	7.2646
10	0.669 832	2847.09	3.8455	7.3976
11	0.667 405	3091.42	3.7835	7.5298
12	0.664 670	3331.14	3.7251	7.6615
13	0.661 674	3566.31	3.6697	7.7930
14	0.658 468	3796.98	3.6172	7.9246
15	0.655 099	4023.19	3.5671	8.0563
16	0.651 618	4245.00	3.5191	8.1885
17	0.648 073	4462.48	3.4731	8.3210
18	0.644 514	4675.67	3.4287	8.4541
19	0.640 988	4884.65	3.3859	8.5878
20	0.637 546	5089.46	3.3443	8.7222
21	0.634 236	5290.17	3.3039	8.8572
22	0.631 108	5486.84	3.2643	8.9930
23	0.628 209	5679.52	3.2256	9.1296
24	0.625 591	5868.25	3.1875	9.2670
25	0.623 300	6053.09	3.1499	9.4052
26	0.621 387	6234.06	3.1126	9.5444
27	0.619 900	6411.22	3.0755	9.6845
28	0.618 889	6584.57	3.0384	9.8257
29	0.618 402	6754.14	3.0012	9.9681
30	0.618 489	6919.94	2.9637	10.1119
31	0.619 198	7081.96	2.9259	10.2570
32	0.620 578	7240.19	2.8875	10.4039

the RKR1 Program of Le Roy.[31] The resultant RKR potential energy curve is listed in pairs of classical turning points in Table IV, which covers only the vibrational range below $v=32$, mainly the outer ionic well. To fully understand various spectral behaviors of the ^{7}LiH $C\,^1\Sigma^+$ state, one should have a potential having an optimum coverage of internuclear distances. To fulfill this end, one may use the ECP potential of Boutalib and Gadéa,[22] our *ab initio* potential, or a hybrid potential,[32] which are shown in Fig. 5. The quality of the two theoretical potentials can be first tested by comparing their resultant rotational energies, B_v, and vibrational energy separations, ΔG_v [defined as $G(v+1)-G(v)$ here], with our observed values. The relevant eigenvalues were numerically calculated by the EIGEN program.[33] For easily visualizing the trend of variation, we compare the B_v values in Fig. 6 and the ΔG_v values in Fig. 7, in which only the vibrational levels above $v=32$ are presented. Note that the observed rotational energy at $v=34$ was estimated from the observed four term values, $T(34,3)$, $T(34,4)$, $T(35,3)$, and $T(35,4)$, assuming that the drifting value due to avoided crossing is the same for each J value. The ΔG_v values are the values for $J=3$ and $J=4$ since the rovibrational levels, (34,3) and (34,4), are among the observed lowest rotational levels of the $v=34$ vibrational level. One can see from Fig. 6(a) that for vibrational levels below $v=27$, the B_v values derived from our *ab initio* potential resemble more the observed values,

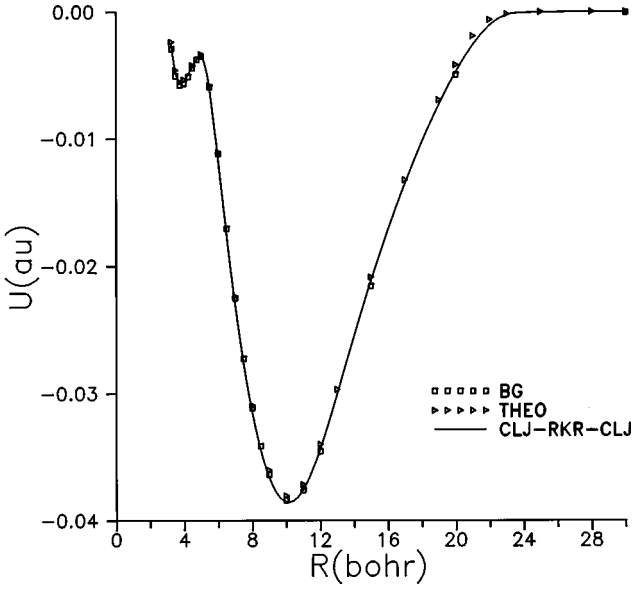


FIG. 5. Potential energy curves of the $C\,^1\Sigma^+$ electronic state. Symbol squares denote the *ab initio* potential of Boutalib and Gadéa (Ref. 22), symbol triangles our *ab initio* potential, and the solid line a hybrid potential, CLJ-RKR-CLJ.

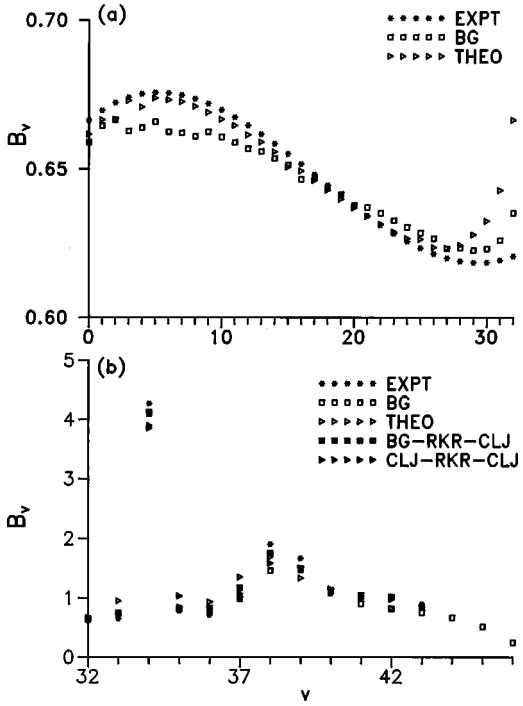


FIG. 6. The rotational energy B_v of the $C\,^1\Sigma^+$ electronic state. Part (a) for the vibrational range of 0–32, part (b) for the vibrational levels beyond $v=33$. In parts (a) and (b), symbol stars denote the observed B_v values; symbol open squares the B_v values of the *ab initio* potential of Boutalib and Gadéa (Ref. 22); and symbol open triangles the B_v values of our *ab initio* potential (see also Table V). In part (b), symbol solid squares denote the B_v values of the BG-RKR-CLJ hybrid potential, and symbol solid triangles the B_v values of the CLJ-RKR-CLJ hybrid potential.

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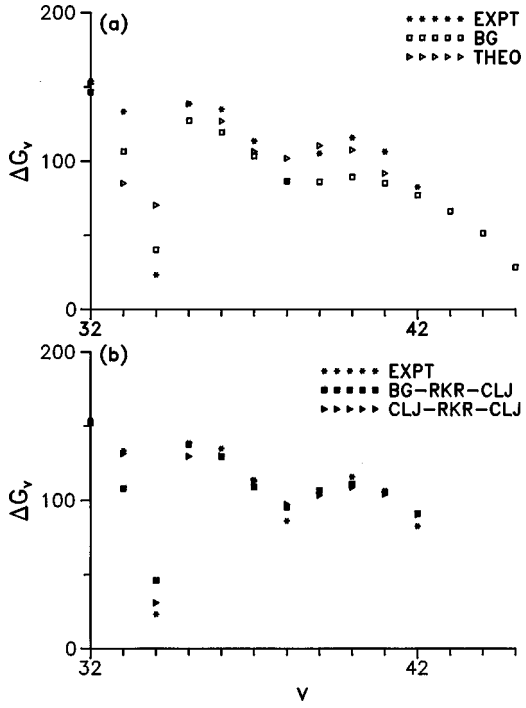


FIG. 7. The vibrational energy separation ΔG_v of the $C\,^1\Sigma^+$ electronic state. Part (a) for the vibrational levels beyond $v=32$. In part (a), symbol stars denote the observed ΔG_v values; symbol open squares the ΔG_v values of the *ab initio* potential of Boutalib and Gadéa (Ref. 22); and symbol open triangles the ΔG_v values of our *ab initio* potential. In part (b), symbol solid squares denote the ΔG_v values for the BG-RKR-CLJ hybrid potential, and symbol solid triangles the ΔG_v values of the CLJ-RKR-CLJ hybrid potential.

and for the levels between $v=29$ and $v=32$, the B_v values derived from the ECP potential of Boutalib and Gadéa[22] resemble more the observed values. This indicates that the lower vibrational levels of the ECP potential of Boutalib and Gadéa[22] lie at longer internuclear distances. The B_v values in Fig. 6(b) and the ΔG_v values in Fig. 7(a) for the vibrational levels above $v=32$ reveal (1) that the shape of the ECP potential of Boutalib and Gadéa[22] is slightly too broad since it sustains 47 vibrational levels, three more than expected, and (2) that our *ab initio* potential sustains only up to the rovibrational level of (43,1). Thus it seems better to find another optimum potential, whose spectroscopic properties may mimic more closely what we have observed.

Two hybrid potentials have been constructed and they are described in Table V. One, denoted as BG-RKR-CLJ, is composed of (region I) a shifted potential for the short internuclear distances, from the ECP potential by Boutalib and Gadéa[22] (region II), our RKR potential listed in Table IV (region III), a scaled potential for the internuclear distances between 23 and 30 bohrs, from our *ab initio* potential, and (region IV) a long-range potential, for the internuclear distances beyond 30 bohr, reported by Proctor and Stwalley.[34] The other, denoted as CLJ-RKR-CLJ and also presented in Fig. 5 as a solid line, is slightly different by replacing a shifted potential for the short internuclear distances from our *ab initio* potential. For the tiny inner well, the location (in bohr) of the potential minimum and that of the potential barrier, and the well depth (in cm^{-1}) relative to the barrier

TABLE V. Hybrid potentials for the ^{7}LiH $C\,^1\Sigma^+$ state. $R_{\min}(32)$ and $R_{\max}(32)$ represent the inner and outer classical turning points of the $v=32$ level. In the inner neutral region I, two *ab initio* potentials were shifted to join our RKR potential listed in Table III; the resultant two hybrid potentials are denoted as BG-RKR-CLJ and CLJ-RKR-CLJ, respectively.

Region	R(bohr)	$U(R)$(a.u.)	Reference
I	$R<R_{\min}(32)$	$U(R)+\Delta U$	[a]1,2
II	$R_{\min}(32)<R<R_{\max}(32)$	RKR potential	This work
III	$R_{\max}(32)<R\leqslant 30$	$\gamma U(R)$	[b]
IV	$R>30$	$-C_6R^{-6}-C_8R^{-8}-C_{10}R^{-10}$	[c]

[a]A shifted *ab initio* potential for the region I. The shifted constant ΔU is (1) −4.628 599 2E-6 for the *ab initio* potential of Boutalib and Gadéa (Ref. 22), and (2) −1.522 628 5E-4 for the *ab initio* potential presented in Table I.

[b]A scaled *ab initio* potential for the region III. The scaled constant γ is 1.099 865 9 for the *ab initio* potential listed in Table I.

[c]A long-range potential of Proctor and Stwalley (Ref. 34). $C_6=512.1$, $C_8=132\,600$, and $C_{10}=409\,700\,00$.

are of 3.825, 4.963, and 516 for BG-RKR-CLJ, and of 3.820, 4.974, and 492 for CLJ-RKR-CLJ. Their rotational energies and vibrational energy separations (both in solid symbols) are indeed close to our observed values, as demonstrated in Figs. 6(b) and 7(b). The quality of these two hybrid potentials can also be assured by comparing their peculiar rovibrational levels with observed values in Table VI, in which the quantum number v_{inner} of the inner vibrational level, the highest rovibrational level of the upmost vibrational level, $(v_{\max},J_{\max})$, their corresponding bound or quasibound characteristics, and term values are compared. The peculiarity of the vibrational level $v=34$ may become visible by portraying its relevant rotation-vibration wave function squares (probability densities), as demonstrated in Fig. 8. The probability densities for the two rovibrational levels, (34,0) and (34,1), are found to concentrate at short internuclear distances, indicating that LiH molecules at these two levels would most often stay inside the inner neutral well. For the range of $v=33-43$, the term values calculated from the CLJ-RKR-CLJ hybrid potential are closer to the observed ones, and they are also presented in Fig. 3.

The observed rovibrational level (43,8) lies at 2.74 cm^{-1} above the dissociation limit and it is a quasibound level. The CLJ-RKR-CLJ hybrid potential predicts that the rovibrational level (43,8) lies at 4.49 cm^{-1} above the dissociation limit and it has a linewidth of $\sim$0.24 cm^{-1}, indicating that it may be observed via a broadened linewidth due to predissociation. However, the linewidths for the observed OODR fluorescence depletion spectral lines are found to vary typically from 0.5 and 1 cm^{-1}, which is larger than the Doppler width of 0.24 cm^{-1} for the $A\,^1\Sigma^+-X\,^1\Sigma^+$ rovibrational transitions,[23] and in turn leads the line broadening to be obscured. It should also be noted that owing to the fact that we failed to observe any LiH $C\,^1\Sigma^+-A\,^1\Sigma^+$ and $C\,^1\Sigma^+-X\,^1\Sigma^+$ resonance fluorescence, we alternatively used the $A\,^1\Sigma^+-X\,^1\Sigma^+$ resonance fluorescence depletion scheme to observe the $C\,^1\Sigma^+$ state. Thus the observed large OODR linewidths may imply that the $C\,^1\Sigma^+$ state is actually a predissociating state due to the nonadiabatical coupling between the $C\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states. Such a dynamical implication

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TABLE VI. Comparison of some peculiar rovibrational levels observed and those from *ab initio* potentials and hybrid potentials for the ^{7}LiH $C\,^1\Sigma^+$ state. v_{inner} is the vibrational quantum number for the vibrational level mainly sustained in the tiny inner well (shell), and ($v_{\max}$,$J_{\max}$) for the highest rotational level of the upmost vibrational level. The superscript * denotes a quasibound level. T_{vJ} is the term value in cm^{-1}.

v_{inner}	($v_{\max}$,$J_{\max}$)	T_{vJ}	Method	Reference
34	(43, 8)*	8468.05	OODR	This work
34	(46, 5)*	⋯	ECP	22
34	(43, 1)*	⋯	*Ab initio*	This work
34	(43, 8)*	8469.7	Hybrid(BG-RKR-CLJ)	This work
34	(43, 8)*	8469.8	Hybrid(CLJ-RKR-CLJ)	This work

will be further investigated, and the results will be presented in a forthcoming paper along with the newly observed $D\,^1\Sigma^+$ state, in which some observed OODR linewidths are found to vary between 0.3 and 2 cm^{-1}.

D. Transition probabilities[32]

In order to understand relevant spectral behaviors of the $C\,^1\Sigma^+$ state and to further confirm the absolute vibrational assignment above, the calculation of transition probabilities among various rotation-vibration bands of the $C\,^1\Sigma^+ - A\,^1\Sigma^+$ electronic system is quite essential. With a given hybrid potential[35] for the $A\,^1\Sigma^+$ state and the hybrid potential, CLJ-RKR-CLJ, and our transition moment function listed in Table I we have calculated some transition matrix elements. They can be compared with the observed optical–optical double resonance fluorescence depletion spectral intensities. To confirm the above vibrational assignment, spectral intensities of several OODR transitions, $C(21\text{-}27,7)$-$A(7,6)$, $C(19\text{-}30,7)$-$A(8,6)$, and $C(22\text{-}32,7)$-$A(9,6)$ were measured, and then they were integrated and normalized to the PROBE laser energy. Table VII lists some calculated transition matrix elements and reveals that three unobserved transitions, $C(23,7)$-$A(7,6)$, $C(20,7)$-$A(8,6)$, and $C(25,7)$-$A(9,6)$, and one weak transition, $C(28,7)$-$A(8,6)$, are each indeed corresponding to a small transition moment. Therefore, the above $v_0=12$ assignment is further confirmed.

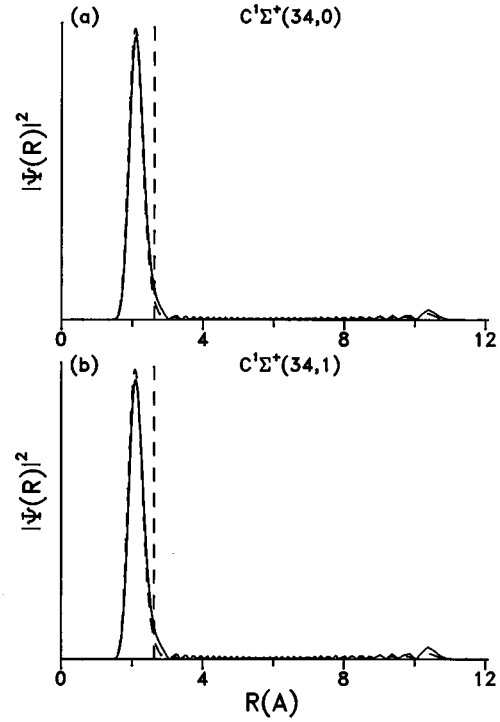


FIG. 8. Probability densities for the two rovibrational levels, (34,0) and (34,1), of the $C\,^1\Sigma^+$ electronic state. The solid and dashed curves represent the probability densities calculated from the CLJ-RKR-CLJ and BG-RKR-CLJ hybrid potentials, respectively. The vertical dashed line denotes the location of the energy barrier between the inner neutral well and the outer ionic well.

TABLE VII. The transition dipole matrix elements [in $10^4(e\text{Å})^2$ units] for some $C\,^1\Sigma^+(v,7) - A\,^1\Sigma^+(v',6)$ transitions, which are most relevant to the vibrational assignment.

v/v'	7	8	9
19	60.44	7.67	21.79
20	42.99	0.05[a]	42.32
21	20.78	9.14	50.49
22	4.31	27.47	41.70
23	0.31[a]	43.43	22.61
24	9.07	48.04	5.50
25	25.04	39.35	0.12[a]
26	39.90	22.65	8.24
27	46.56	6.95	23.54
28	42.17	0.05[a]	36.12
29	29.04	4.66	38.44
30	13.34	17.46	29.42
31	2.24	31.01	14.56
32	0.69	37.62	2.63

[a]Unobserved transitions.
[b]Weak transition.

V. CONCLUSIONS

In sum, 42 vibrational ($v=2-43$) levels of the ^{7}LiH $C\,^1\Sigma^+$ excited electronic state have been observed by a pulsed optical–optical double resonance fluorescence depletion spectroscopic technique. The absolute vibrational numbering of the $C\,^1\Sigma^+$ state is identified with the measurements of the isotopic shifts between ^{7}LiH and ^{6}LiH among several rotation-vibration bands of the $C\,^1\Sigma^+ - A\,^1\Sigma^+$ electronic system, and it is further demonstrated by the consistency between observed spectral intensities and calculated ones. The highest observed quasibound rotational level for the $v=43$ level is $J=8$. The spectral term values for those vibrational levels lying above $v=33$ occur in an irregular order. The level $v=34$ lies mainly in the inner, tiny well. The term values for the observed rovibrational levels ($2\leqslant v\leqslant 32$, $0\leqslant J\leqslant 11$) in the outer ionic well are described by a set of Dunham-type coefficients, with which an RKR potential energy curve is constructed. This RKR potential is combined with our *ab initio* potential to construct an effective hybrid potential. The calculated electronic transition moment function for the C-A transition is also given. Our *ab initio* calculation also reveals that the potential energy curve for the higher $D\,^1\Sigma^+$ state has a more complex form, and the experimental work on this state is currently in progress.

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